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To cite this article: Chen Guang-Hua & Huang Rong-Hui (2008) Role of Equatorial Wave Transitions in Tropical Cyclogenesis over the Western North Pacific, Atmospheric and Oceanic Science Letters, 1:1, 64-68, DOI: [10.1080/16742834.2008.11446765](https://doi.org/10.1080/16742834.2008.11446765)

To link to this article: <https://doi.org/10.1080/16742834.2008.11446765>



Published online: 03 Jul 2015.



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Role of Equatorial Wave Transitions in Tropical Cyclogenesis over the Western North Pacific

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Received 19 June 2008; revised 11 August 2008; accepted 11 August 2008; published 16 November 2008

Abstract This study associates tropical cyclone (TC) activity over the western North Pacific (WNP) with the equatorial wave transition from an interannual viewpoint, revealing that the tropical cyclogenesis mean location may be modulated by a longitudinal shift in the transition of Mixed Rossby-gravity (MRG) waves to off-equatorial tropical depression (TD) disturbances from year to year. To a large extent, the wave transition is attributable to the monsoon trough in response to the thermal state of the warm pool (WP) over the WNP. During the cold state years in the WP, the basic flow confluence region associated with the monsoon trough penetrates eastward, leading to an eastward shift in the location of the wave transition. Such an environment, in which wave accumulation and energy conversion occur, is favorable for tropical cyclogenesis; as a result, the averaged cyclogenesis location moves eastward. The condition is reserved during the warm years in the WP, resulting in the prominent westward-retreating mean TC formation.

Keywords: equatorial waves, tropical cyclogenesis, western North Pacific

Citation: Chen, G. H., and R. H. Huang, 2008: Role of equatorial wave transitions in tropical cyclogenesis over the western north Pacific, *Atmos. and Oceanic Sci. Lett.*, **1**, 64–68.

1 Introduction

A great deal of research has been carried out investigating the interannual variation of tropical cyclone (TC) activity and its related large-scale circulation. The consensus has been that there is a significant increase in the TC activity over the southeast portion of the western North Pacific (WNP) and a decrease in the northwest quadrant during strong El Niño years, which is associated with the southeastward shift of the monsoon trough. During strong La Niña years, the conditions are reversed (Chen et al., 1998; Chan, 2000; Wang and Chan, 2002). In addition to El Niño–Southern Oscillation (ENSO) events, recent research has found that non-local atmospheric modes, such as the North Pacific oscillation (NPO), Antarctic oscillation (AAO), and Asian-Pacific oscillation (APO), can have an important influence on TC activity over the WNP through teleconnection wave-like patterns (Ho et al., 2005; Wang et al., 2007; Wang and Fan, 2007; Fan, 2007a; Zhou et al., 2008). Furthermore, it becomes feasible to quantitatively extract the influential signals,

which act as predictors of TC frequency (Fan, 2007b). These large-scale circulation modes only provide favorable backgrounds for cyclogenesis, while the fundamental triggering factors should be attributed to tropical synoptic disturbances that have a temporal and spatial scale similar to TCs (Frank and Roundy, 2006). However, the mechanism by which tropical cyclogenesis is modulated by synoptic-scale disturbances from the interannual viewpoint is not fully understood.

Evidence is growing that equatorial waves play an important role in tropical cyclogenesis in the WNP (Takayabu and Nitta, 1993; Chang et al., 1996; Frank and Roundy, 2006). A number of studies have found a strong link between TC formation and equatorial wave transformation, the latter essentially referring to the transition of Mixed Rossby-gravity (MRG) waves to off-equatorial tropical depression (TD) type disturbances (Dickinson and Molinari, 2002). Earlier studies revealed that once the equatorial westward-propagating MRG waves originating from the eastern and central Pacific penetrate into the WNP monsoon region, they can transit to northwestward-propagating TD-type disturbances accompanied by a gradual change in wave number, phase speed, and propagation direction (Lau and Lau, 1990; Liebmann and Hendon, 1990).

In addition, Chen and Huang (2008) have showed that the thermal state of the warm pool (WP) in the WNP region could be well correlated with tropical convection, monsoon circulation, and tropical wave features. Therefore, the present study is an attempt to provide explanations for the interannual variations in tropical wave transition over the WNP and its impacts on tropical cyclogenesis as a way to clarify the linkage between tropical waves and the WP thermal states.

2 Data and methodology

The study employs daily National Center for Environmental Prediction (NCEP) reanalysis variables and TC datasets from the Joint Typhoon Warning Center (JTWC). Daily outgoing long wave radiation (OLR) derived from the National Oceanic and Atmospheric Administration–Cooperative Institute for Research in Environmental Sciences (NOAA-CIRES) Climate Diagnostics Center is used as a proxy for deep tropical convection. The subsurface ocean temperature data is archived from the Joint Environmental Data Analysis Center. The present study focuses on the period from July to October during the years of 1980–2003.

Using a time Butterworth filter, each field was filtered to extract a 3–6-day cutoff period in order to isolate the tropical disturbance of interest. A linear regression was performed in order to examine the propagation and transition of tropical waves. Following the method advanced by Bretherton et al. (1999), the effective sample size (ESS) is re-derived for a correlation significance test.

3 Results and discussion

The WP thermal state is represented by an area-mean subsurface ocean temperature anomaly (SOTA) at 120 m depth over the region of 0–16°N and 125–165°E. The subsurface ocean temperature over the WNP is used rather than sea surface temperature because SOTA signals are easily detected due to their large amplitude. Warm (cold) years are defined when this index is above 1.0°C (below –1.0°C). In order to exclude the influence of the ENSO signal to isolate the WP effect, the years in which the SST anomalies in the Niño 3.4 region are out of the range of $\pm 1.0^\circ\text{C}$ during the months of interest should be ruled out. Therefore, there are three warm state years (1983, 1996, and 2000) and three cold state years (1980, 1986, and 1994).

The 850 hPa streamline was composited for the warm and cold years. During the warm years, the monsoon trough retreats westward and northward (Fig. 1a), so that more TCs tend to form in the northwest region of the WNP. In contrast, during the cold years, westerly anomalies in the western equatorial Pacific extend to the dateline and even beyond (Fig. 1b). These westerly anomalies encounter trade easterly winds and enhance the cyclonic shear so that the monsoon trough is located at a lower

latitude and penetrates further eastward, which results in more TCs forming in the southeast quadrant of the WNP.

What role do the westward-propagating equatorial waves play in tropical cyclogenesis? To answer this question, the composites of filtered wind at 850 hPa and the OLR field with respect to TC genesis locations were computed. The TCs forming equatorward of 20°N from 120°E to 180°E, a dominant region of the monsoon trough, are involved in these composites. Figure 2a shows the composite for three days prior to cyclogenesis during the cold years. A circulation characterized by alternating cyclonic and anticyclonic structures appears near the cyclogenesis location, which is centered in the domain. The enhanced (suppressed) convection anomaly is located in the center of cyclonic (anticyclonic) circulation. Meanwhile, a weak cyclonic flow, anchored approximately 4000 km to the southeast of the TC genesis location (12.3°N, 146.0°E), originated from the equator over the central Pacific. This weak cyclonic vorticity (labeled X) gradually moves westward and intensifies, which is depicted in Fig. 2b. By two days prior to cyclogenesis, the cyclonic disturbance X shifts horizontally and eastward about 1500 km, accompanied by the anomalous convection to the northeast. The structure of anomalous convection, which lags near the cyclonic center by one quarter of wavelength, is similar to that of equatorial MRG waves (Takayabu and Nitta, 1993; Dickinson and Molinari, 2002; Frank and Roundy, 2006). By the lag of –1 day, the cyclonic disturbance X has propagated northwestward, which is close to the center (Fig. 2c). Moreover, enhanced convection is superposed over the center of cyclonic circulation with a southwest-northeast tilt. The dissimilarities in wave structures show that a transition between two wave types has occurred one day prior to cyclogenesis. At the time of cyclogenesis, the composited TCs genesis location is in superposition with cyclonic center, accompanied by wavetrain-like structures that are coupled with convection (Fig. 2d). Based on these composites, it could be inferred that a large fraction of TCs genesis are associated with the transitions of two wave types, which are likely to occur due to eastward and equatorward penetration of the monsoon trough. Similarly, the composite patterns for TC during the warm years resemble the counterpart during the cold years, except for the relatively obscure wave-like structure that appears as a result of the weakening of monsoon trough (not shown).

To show the longitudinal shift of wave transition corresponding to different years, linear regressions are calculated for wind and OLR with respect to meridional wind at the base grid point (0°, 160°E) during the warm and cold year. The field regressions are implemented for one leading day, one simultaneous day and one lagging day with respect to the base point. For the warm years (Fig. 3a), at lag –1 day, a cyclone center is located in the base point (0°, 160°E). This cyclone gradually propagates westward, and is replaced by an anticyclone at lag +1 day. The regression pattern is symmetrical about the equator for meridional wind and anti-symmetrical for zonal winds

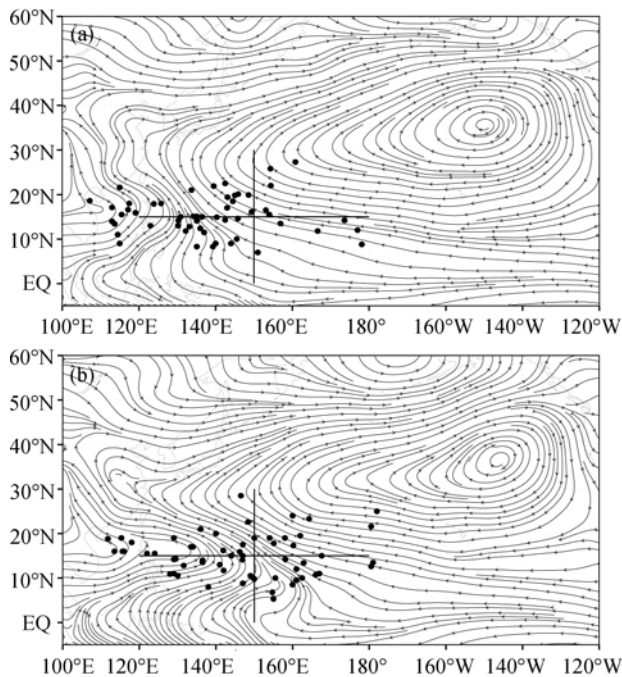


Figure 1 850 hPa composited stream field and TC genesis location denoted by black dots (a) during the warm years; (b) during the cold years. For ease of comparison, the 15°N and 150°E lines are plotted.

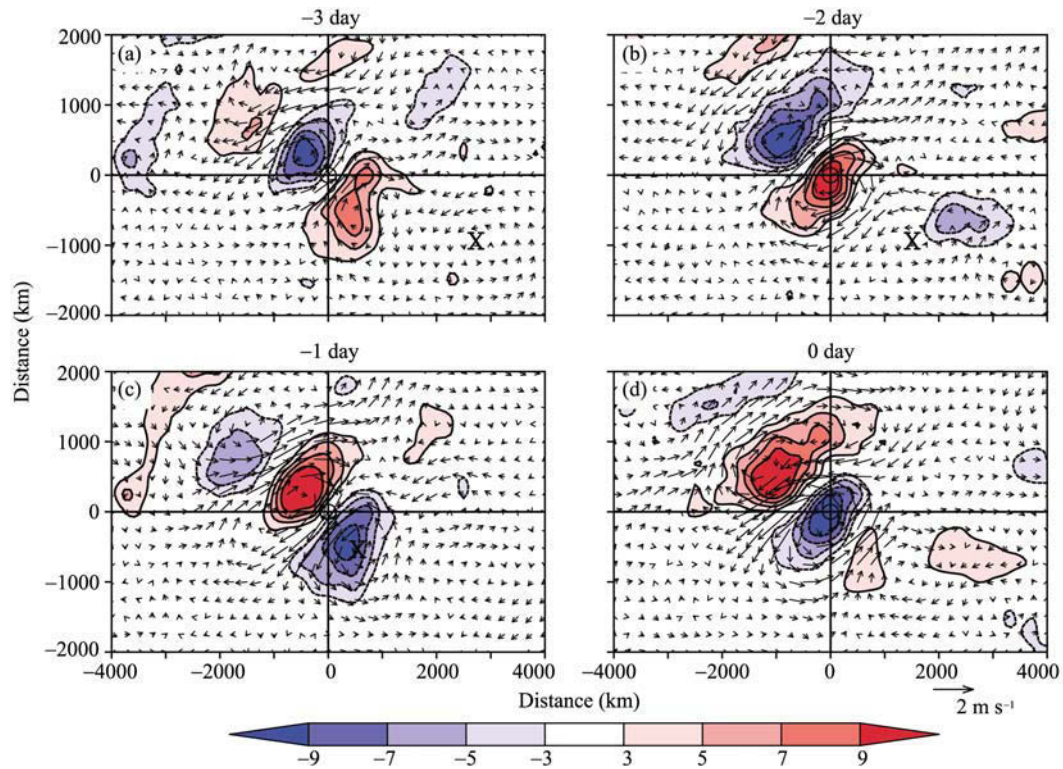


Figure 2 Composites of the filtered vector winds at 850 hPa and OLR (shading) relative to the mean genesis location of 70 TCs during the cold years for (a) 3 days, (b) 2 days, (c) 1 day and (d) 0 day prior to cyclogenesis.

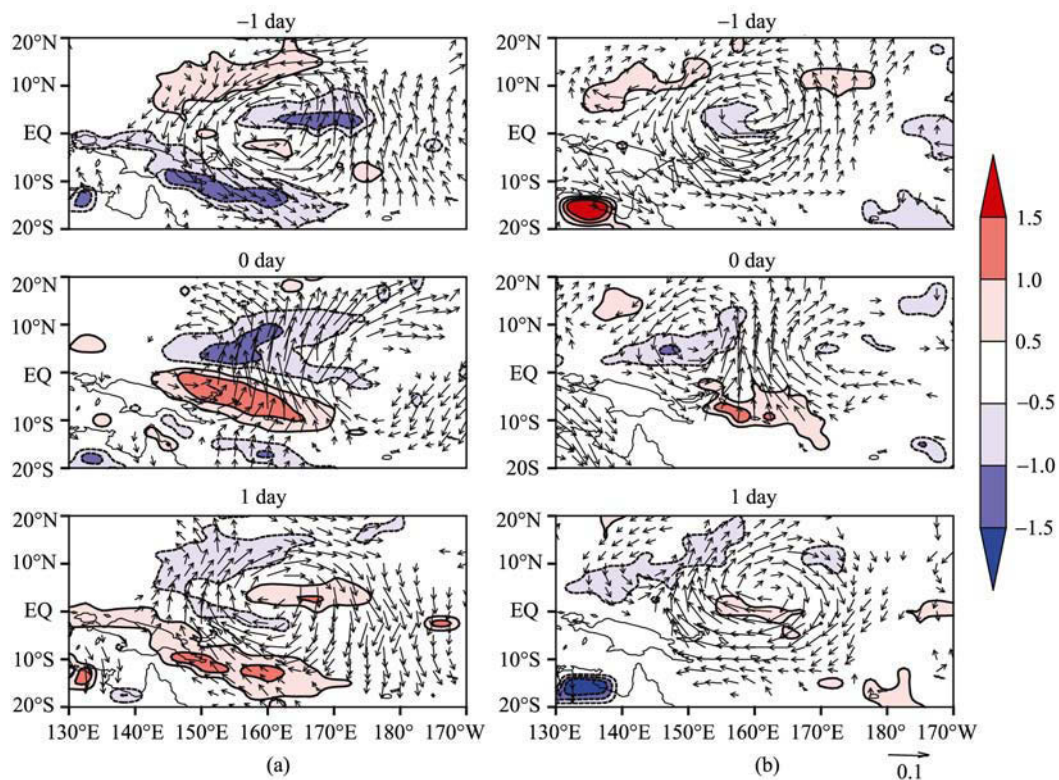


Figure 3 Linear regression of 850 hPa winds (vector, $m s^{-1}$) and OLR (shading, $W m^{-2}$) with respect to 850 hPa meridional wind at the base point ($0^\circ, 160^\circ E$), derived from 3–6-day filtered data for successive three lag-lead days during (a) the warm years and (b) the cold years. Wind vectors and OLR are plotted only when the OLR regression and either the u or v components exceed a 95% confidence level.

and OLR, which is representative of the MRG wave structure. In addition, the anomalous convective center is not superposed over the cyclonic center, but located to the northeast of the cyclonic center and northwest of the anticyclonic center. This feature resembles what Takayabu and Nitta (1993) depicted for MRG waves that are “loosely” coupled with convection, and indicates the survival of MRG wave structures around 160°E and the westward-shift of transition zone of MRG-TD waves.

In contrast, a different pattern during the cold years is presented in Fig. 3b. The cyclonic center departs from the equator and is “tightly” coupled with convection (i.e. the cyclonic center almost coincides with the active convection region). A hybrid structure of MRG waves and TD disturbances is seen to the east of 160°E. The off-equatorial waves move northwestwards and evolve gradually into TD-type disturbances, whose wavelength becomes shortened. The major axes of these disturbances are oriented along a northeast-southwest line, which indicates that there is a conversion from mean kinetic energy to eddy kinetic energy (Kwon and Frank, 2005). The distinctive features between the two kinds of years imply that the monsoon trough in response to the WP thermal state has a large impact on the longitudinal location of wave transition. The transition zone shifts westward in the warmer years compared to that in the colder years.

The mechanism for the wave transition and its influence on cyclogenesis can be interpreted as follows. Considering the zonal wavenumber change following a ray in a time-invariant mean flow, the formula $dk/dt = -k (d\bar{u}/dx)$, derived by Webster and Chang (1988), is still applicable to MRG waves. It means that the opposite zonal flows of the convergence zone southeast of monsoon trough could cause westward-propagating waves to produce a zonal-scale compression and accumulation of energy (Chang and Webster, 1990; Holland, 1995). The zonal scale compression means that there is an increase in the wave-number and a reduction in the wavelength as the waves enter the convergence zone. From the schematics for theoretical wavenumber-frequency domain of equatorial waves (Matsuno, 1966), it is clear that the MRG wave regime asymptotes to the Rossby wave regime follow the increase in the wave-number. Hence, there is a possible interpretation for the transition of MRG waves to TD-type disturbances with Rossby wave properties in the convergence zone of zonal flow.

Using a simple numerical simulation, Aiyer and Molinari (2003) examined the evolution of MRG waves in a monsoon trough environment in which the trough has a northwest-southeast orientation and increased low-level convergence in the eastern edge. In their study, the westward-moving equatorially trapped MRG wave contracts as it encounters the region of low-level convergence, and then, a secondary wave pattern develops along the monsoon trough axis due to a wave-mean flow interaction such that energy is transferred from the basic state to the wave. Additionally, the effect of the easterly vertical shear of the mean flow serves as an important internal dynamic factor that drives Rossby waves moving northward, a hy-

pothesis that has been substantiated by numerical experiments (Wang and Xie, 1997). Therefore, wave accumulation and energy conversion from basic monsoonal flow to eddy waves during the process of wave transition concurrently contribute to favorable conditions for cyclogenesis.

4 Summary

The present analysis examined a close correlation between longitudinal variation in the low tropospheric wave transition zone and the monsoon trough in response to the thermal state in the WP. When the WP is in the warm phase, the monsoon trough lies immediately to the east of Philippine islands. This induces a westward shift of the wave transition zone. In contrast, during the cold phase of the WP, the active convection region and the monsoon trough extend eastward to the central Pacific. This prohibits the MRG waves from moving westward into the western portion of the WNP due to eastward displacement of wave transition zone. Additionally, it is shown that the evolution of MRG waves into off-equatorial disturbances is conducive to cyclogenesis through wave accumulation and kinetic energy conversion arising from the monsoonal flow confluence and cyclonic shear. The results suggest that the interannual variation in the mean location of TC formation is, to a certain degree, attributable to longitudinal shifts in the transition zone of tropical waves. However, it is worthwhile to mention that, due to the monsoon trough that penetrates eastward and equatorward during the cold years, this mechanism for wave-induced cyclogenesis may be more significant for the cold years than for the warm years.

Acknowledgments. This study is supported by Project KZCX2-YW-220, Program of Knowledge Innovation for the 3rd Period, Chinese Academy of Sciences; Project 40730952, the National Natural Science Foundation of China, and the Project G2006 CB403600, the “National Key Program for Developing Basic Sciences” respectively.

References

- Aiyer, A., and J. Molinari, 2003: Evolution of mixed Rossby-gravity waves in idealized MJO environments, *J. Atmos. Sci.*, **60**, 2837–2855.
- Bretherton, C. S., M. Widmann, V. P. Dymnikov, et al., 1999: The effective number of spatial degrees of freedom of a time-varying field, *J. Climate*, **12**, 1990–2009.
- Chan, J. C. L., 2000: Tropical cyclone activity over the western North Pacific associated with El Niño and La Niña Events, *J. Climate*, **13**, 2960–2972.
- Chang, C. -P., J. M. Chen, P. A. Harr, et al., 1996: Northwestward propagating wave patterns over the tropical western North Pacific during summer, *Mon. Wea. Rev.*, **124**, 2245–2266.
- Chang, H. R., and P. J. Webster, 1990: Energy accumulation and emanation at low latitudes. Part II: Nonlinear response to strong episodic equatorial forcing, *J. Atmos. Sci.*, **47**, 2624–2644.
- Chen, G. H., and R. H. Huang, 2008: Influence of monsoon over the warm pool on interannual variation of tropical cyclone activity over the western North Pacific, *Adv. Atmos. Sci.*, **25**(2), 319–328.
- Chen, T. C., S. P. Weng, N. Yamazaki, et al., 1998: Interannual variation in the tropical cyclone formation over the western North Pacific, *Mon. Wea. Rev.*, **126**, 1080–1090.

- Dickinson, M., and J. Molinari, 2002: Mixed Rossby-gravity waves and western Pacific tropical cyclogenesis. Part I: Synoptic evolution, *J. Atmos. Sci.*, **59**, 2183–2196.
- Fan, K., 2007a: Sea ice cover over North Pacific, a predictor for the typhoon frequency over west North Pacific? *Science in China (D)*, **50**(8), 1251–1257.
- Fan, K., 2007b: New predictors and a new prediction model for the typhoon frequency over western North Pacific, *Science in China (D)*, **50**(9), 1417–1423.
- Frank, W. M., and P. E. Roundy, 2006: The role of Tropical waves in tropical cyclogenesis, *Mon. Wea. Rev.*, **134**, 2397–2417.
- Ho, C. H., J. H. Kim, H. S. Kim, et al., 2005: Possible influence of the Antarctic Oscillation on tropical cyclone activity in the western North Pacific, *J. Geophys. Res.*, **110**, D19104, doi: 10.1029/2005JD005766.
- Holland, G. J., 1995: Scale interaction in the western Pacific monsoon, *Meteor. Atmos. Phys.*, **56**, 57–79.
- Kwon, Y., and W. M. Frank, 2005: Dynamic instabilities of simulated hurricane-like vortices and their impacts on the core structure of hurricanes. Part I: Dry experiments, *J. Atmos. Sci.*, **62**, 3955–3973.
- Lau, K. H., and N. G. Lau, 1990: Observed structure and propagation characteristics of tropical summertime synoptic-scale disturbances, *Mon. Wea. Rev.*, **118**, 1888–1913.
- Liebmann, N., and H. H. Hendon, 1990: Synoptic-scale disturbances near the equator, *J. Atmos. Sci.*, **47**, 1463–1479.
- Matsuno, T., 1966: Quasi-geostrophic motions in the equatorial area, *J. Meteor. Soc. Japan*, **44**, 25–43.
- Takayabu, Y. N., and T. Nitta, 1993: 3–5 day period disturbances coupled with convection over the tropical Pacific Ocean, *J. Meteor. Soc. Japan*, **71**, 221–245.
- Wang, B., and J. C. L. Chan, 2002: How strong ENSO events affect tropical storm activity over the western North Pacific, *J. Climate*, **15**, 1643–1658.
- Wang, B., and X. S. Xie, 1997: A model for the boreal summer intraseasonal oscillation, *J. Atmos. Sci.*, **54**, 72–86.
- Wang, H. J., J. Q. Sun, and K. Fan, 2007: Relationships between the North Pacific oscillation and the typhoon/hurricane frequencies, *Science in China (D)*, **50**(9), 1409–1416.
- Wang, H. J., and K. Fan, 2007: Relationship between the Antarctic oscillation and the western North Pacific typhoon frequency, *Chinese Science Bulletin*, **52**(4), 561–565.
- Webster, P. J., and H. R. Chang, 1988: Energy accumulation and emanation region at low latitudes: Impacts of a zonally varying basic state, *J. Atmos. Sci.*, **45**, 803–829.
- Zhou, B. T., X. Cui, and P. Zhao, 2008: Relationship between the Asian-Pacific oscillation and the tropical cyclone frequency in the western North Pacific, *Science in China (D)*, **51**(3), 380–385.